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PERSPECTIVE

From application-first to science-first: A two-phase validation framework for complex *in vitro* NAMs

Yiguang Zhu[1,*] and Paul A. Locke[1]

Validation of new approach methodologies increasingly requires developers to define a context of use (CoU) before establishing scientific credibility, constraining complex *in vitro* models. This CoU first strategy can impose a structural limitation on validation. We propose a two-phase framework that separates scientific validation from context-specific qualification, bridged by a shared data repository. Under this structure, the model's scientific capabilities and limitations inform the range of potential applications, rather than being constricted to the first CoU selected.

New approach methodologies (NAMs) promise more human-relevant evidence in toxicity testing and drug development. Their regulatory adoption depends on validation that demonstrates reliability, reproducibility, relevance, and predictability. Federal agencies therefore play a central role in defining validation pathways that build confidence while supporting future innovation. Yet NAM validation increasingly begins with context of use (CoU) definition, asking how a method fits one specific purpose rather than considering its potential as a whole. For complex *in vitro* NAMs such as microphysiological systems (MPS), constricting the CoU can channel validation into a narrow scope before the model's broader capabilities have been explored, which may slow appropriate adoption and delay its benefit for other uses.

A 2023 industry survey documents this pattern: 80% of pharmaceutical companies use MPS internally to guide candidate selection, yet most exclude MPS data from regulatory submissions due to uncertainty over how agencies will receive them.[1] Evidence from human-relevant models is generated but rarely reaches submission. This perspective attributes the gap in part to the CoU-first validation strategy and proposes a robust framework that decouples scientific validation from context-specific qualification.

HOW CoU-FIRST VALIDATION BECAME DOMINANT

US regulatory agencies have increasingly organized NAM validation around CoU and fit-for-purpose logic. The US Food and Drug Administration (FDA) established CoU as central to integrating emerging methods into safety assessment in its 2017 Predictive Toxicology Roadmap.[2] FDA's 2020 Drug Development Tool (DDT) guidance formalized a CoU-based qualification process.[3] Within the DDT guidance, the Innovative Science and Technology Approaches for New Drugs (ISTAND) program is designed to qualify novel tools like complex NAMs.[4] FDA's March 2026 draft guidance on the use of NAMs in drug development continues this trend, organizing validation around CoU, human biological relevance, technical characterization, and fit-for-purpose.[5] The Interagency Coordinating Committee on the Validation of Alternative Methods (ICCVAM) presented a cross-agency framework for NAM validation in its 2024 report, making CoU-based strategy the dominant organizing logic across US regulatory practice.[6]

WHY COMPLEX *IN VITRO* NAMs REQUIRE A DIFFERENT APPROACH

Conventional NAMs, such as skin, ocular, and phototoxicity assays, benefit from mature reference data, established benchmarking strategies, and clear regulatory pathways. Complex NAMs differ in two respects. First, they are platform-, material-, and protocol-dependent: their outputs reflect not only underlying biology but also cell sourcing, culture conditions, device materials, flow, mechanical cues, and analytical pipelines.[7] Second, because human *in vivo* comparison data are scarce, complex NAMs are often benchmarked against animal studies, introducing double extrapolation when NAM findings are first interpreted through animal data and then inferred back to human biology.[8]

The FDA Modernization Act 2.0 broadened the statutory language for nonclinical

[1]Department of Environmental Health and Engineering, Johns Hopkins University, Bloomberg School of Public Health and Whiting School of Engineering, Baltimore, Maryland, USA. *Correspondence: Yiguang Zhu (yzhu99@jh.edu)

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testing, formally recognizing the acceptability of human biology-based methods alongside animal studies.[9] Yet this statutory opening has not yet translated into routine regulatory reliance on complex NAMs. Several complex NAMs have entered ISTAND since 2024, but none has completed full qualification.[4] NAM data continue to enter submissions mainly as supplemental evidence, and reported cases of NDA or IND filings supported substantially by NAM data remain exceptions.[10] This gap reflects several interacting factors, including standardization, cost, and stakeholder confidence; the CoU-first strategy is the factor we examine here.

THREE LIMITATIONS OF CoU-FIRST VALIDATION

The ICCVAM framework captures both the strengths and limits of CoU-first validation. As illustrated in **Figure 1a**, validation begins with a particularized CoU definition, followed by biological relevance assessment, technical characterization,

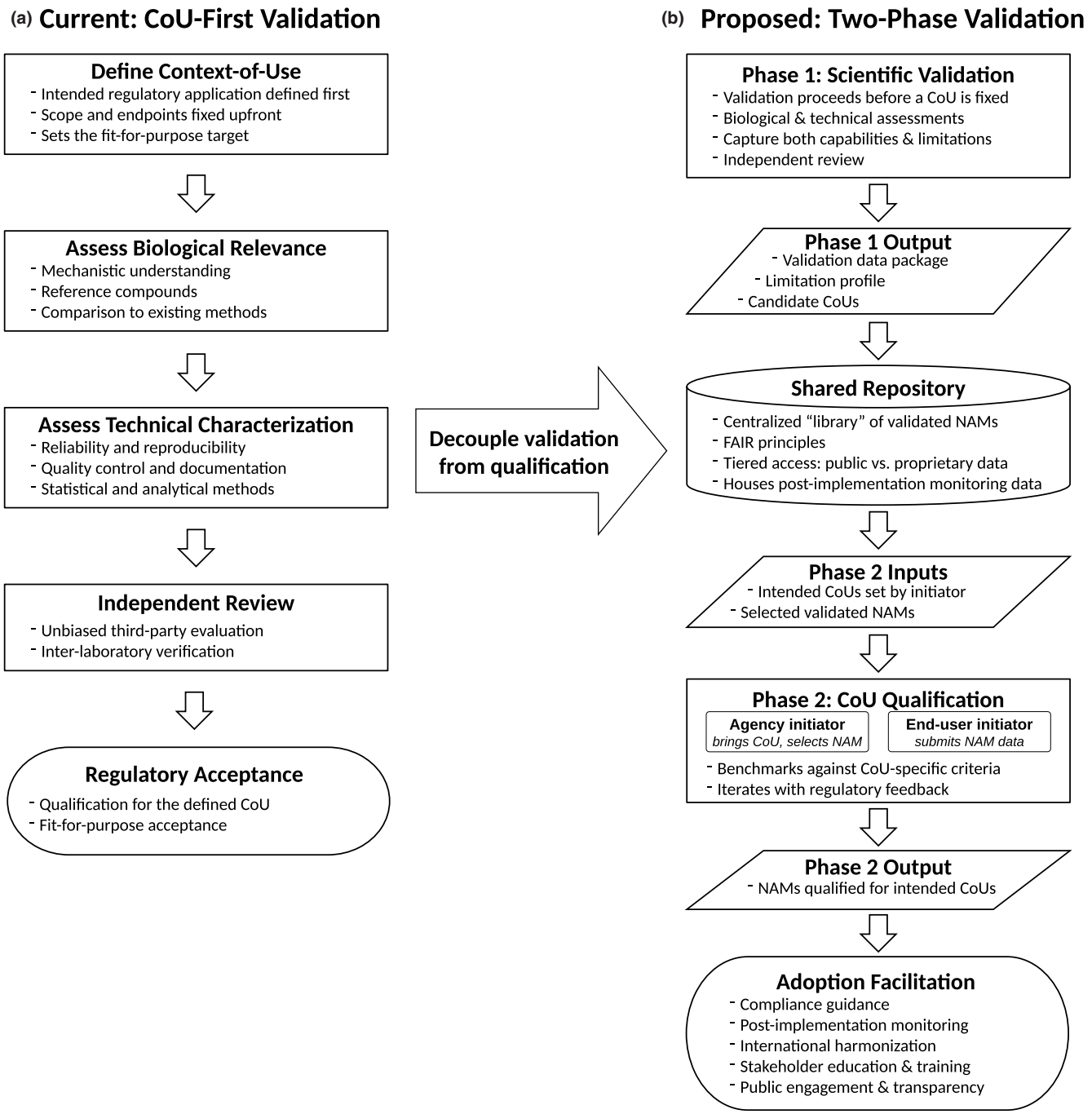


Figure 1 Current and proposed frameworks for validating complex *in vitro* NAMs. (a) Context of use (CoU)-first validation fixes the CoU at the outset and organizes biological relevance, technical characterization, and independent review around that single application before regulatory acceptance. (b) The proposed two-phase framework decouples scientific validation (phase I) from CoU qualification (phase II) via a shared repository of validated new approach methodologies (NAMs), enabling one validated NAM to support multiple CoU.

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independent review, and regulatory acceptance. The framework provides a useful foundation for identifying the major domains of validation. The limitation lies in when CoU is defined, not in what the framework assesses. For complex NAMs, this framework creates three problems.

Structural rigidity

The sequence assumes that a CoU can be defined before a model is fully characterized. For complex NAMs, however, their capabilities, limitations, and appropriate applications often emerge during iteration. Once a CoU is fixed, developers are incentivized to optimize the platform and experimental design around that initial context. Validation then becomes a test against the intended CoU rather than an assessment of the fidelity of the model. The first CoU filed can potentially narrow the evidentiary pathway before a NAM's broader applications are explored. Consider a cardiac MPS first submitted for proarrhythmic risk screening. It would likely be validated around CiPA-aligned endpoints, even though the same platform could support broader cardiotoxicity or disease modeling applications.

Operational challenges

The framework identifies what should be assessed, but not who is responsible for each stage, creating pathway fragmentation. Without a shared validation record or coordinating authority, a single model may require redundant validation across agencies, divisions, or programs. Stakeholder collaboration also remains unstructured: agencies encourage early engagement but provide no clear process for resolving questions about data interpretation, evidentiary sufficiency, or CoU boundaries. The burden of navigating ambiguity falls disproportionately on developers.

Technical limitations

With a predetermined CoU, validation tends to prioritize endpoints that are easy to benchmark and most defensible at the expense of human-mechanistic readouts that make complex NAMs valuable. The same cardiac MPS discussed above would likely be evaluated around endpoints such as action potential duration, repolarization triangulation, or early afterdepolarizations, because these measures are readily benchmarked against existing CiPA datasets. Yet endpoints that distinguish cardiac MPS from monolayer assays, including mechano-electric coupling, tissue-level conduction, and disease phenotype modeling, would remain secondary despite their distinctive human-relevant value.

A TWO-PHASE FRAMEWORK

The proposed framework, illustrated in **Figure 1b**, separates scientific validation (phase I) from CoU-specific qualification (phase II), bridged by a shared repository. The decoupling shifts CoU definition from the entry requirement to an output of scientific characterization that guides phase II qualification. The framework also defines stakeholder responsibilities and prioritizes human mechanistic endpoints over generic readouts. **Table 1** summarizes

Table 1 Feasibility and implementation considerations for the proposed two-phase framework

Consideration	How the proposed framework addresses it
Data ownership and access	The repository uses tiered FAIR access. Public-facing elements include nonconfidential metadata, SOP summaries, performance metrics, limitation profiles, and candidate CoUs. Controlled-access elements include raw data, proprietary analytical pipelines, engineering specifications, and donor-specific materials. Developers retain ownership of proprietary data and intellectual property.
Coordination and roles	A coordinating body (ICCVAM or equivalent) provides scientific stewardship by coordinating repository standards, benchmark datasets, evaluation criteria, and independent review; developers design experiments, generate data, and document protocols and SOPs; independent reviewers (e.g., CROs) provide unbiased verification; regulatory agencies retain all CoU-specific qualification and acceptance decisions in phase II.
Funding and collaboration	Phase 1 validation and repository maintenance are resource intensive and would require support from funding bodies, developers, industry end users, and public–private consortia. The repository lets developers and potential end users benchmark platforms and reuse nonconfidential validation evidence, reducing duplicative effort while protecting proprietary information.
Qualification pathways and CoU adaptation	The qualifying body is the relevant regulatory agency. One validation package can support qualification across multiple CoUs; where it does not fully address a specific CoU, the agency may request supplemental evidence, developed iteratively with the developers and end users best positioned to provide it.
End users in the exploratory phase	Because scientific credibility is established in phase I, exploratory use requires no revalidation. End users may apply a phase I-validated NAM for internal decision making, with local verification as needed, and later draw on the resulting evidence for a phase II filing, a graduated path from internal use to formal qualification.
Postimplementation monitoring	Qualified NAMs would undergo continued monitoring of quality control, SOP adherence, reproducibility, and performance drift. End users and developers may contribute performance and refinement data to the repository, allowing limitation profiles, candidate CoUs, and evaluation criteria to be updated over time.
International harmonization	The framework supports international harmonization by structuring phase I data packages around interoperable validation elements, shared terminology, and reporting formats aligned with OECD and ICH guidance and other jurisdictional regulatory expectations. These data packages could inform future international guideline development while leaving qualification decisions to individual regulators.

Abbreviations: ICH, International Council for Harmonization of Technical Requirements for Pharmaceuticals for Human Use; OECD, Organization for Economic Co-operation and Development.

how the framework addresses key implementation considerations.

Phase I: Scientific validation

Phase I establishes a NAM's scientific credibility and technical robustness before it is qualified for a specific regulatory application. It centers on two assessments drawn from the ICCVAM framework but repositioned earlier in the sequence. Biological relevance assesses mechanistic fidelity to human biology, benchmarks responses against reference compounds, and compares outputs against *in vitro*, animal, and clinical evidence. For a cardiac MPS, this would include cellular composition, tissue organization, electrophysiological and contractile function, integrative readouts such as mechano-electric coupling, and pharmacological responsiveness. Technical characterization then assesses whether those outputs are reproducible and reliable under defined conditions via intra- and interlaboratory studies, supported by design specifications, SOPs, and analytical pipelines. Here, acceptance criteria are set from reproducibility and reliability thresholds rather than CoU-specific performance targets, which are established later in qualification. Independent reviewers evaluate submitted data to reduce bias.

The output of phase I is a validation package that includes performance data, engineering and design specifications, SOPs, analytical methods, a limitation profile, and candidate CoUs. Assembling this package will be a shared effort, and the framework assigns stakeholder responsibilities at entry rather than negotiating them ad hoc across overlapping pathways, reducing the fragmented, redundant review that burdens developers under current practice. **Table 1** details each stakeholder's role.

Shared repository

The repository is the structural innovation that connects the two phases, letting one validation effort serve multiple downstream applications. After phase I, the validation package is deposited and made accessible to agencies and the broader stakeholder community. The repository follows FAIR principles (Findable, Accessible, Interoperable, Reusable) with tiered access to balance transparency and proprietary protection. It is also a living system that supports postimplementation monitoring as models evolve. **Table 1** outlines the data-access, coordination, and monitoring considerations this structure entails.

Phase II: CoU qualification

Qualification determines whether a scientifically validated NAM is fit for a defined CoU. Phase II can be initiated through two pathways. An agency may select an NAM from the repository based on a specific regulatory need, define the intended CoU, and initiate formal qualification. Alternatively, an end user may apply a phase I-validated NAM within its candidate CoUs and submit NAM-generated data in a regulatory filing, providing case-specific evidence. Regardless of the initiation pathway, the qualifying body is the relevant regulatory agency, which evaluates CoU-specific performance criteria such as sensitivity and specificity and assesses operational feasibility, such as cost, throughput, and scalability. Where the phase I validation package does not fully support a CoU, the qualifying body may request supplemental evidence (**Table 1**).

A cardiac MPS in the shared repository could be selected for phase II qualification through either initiation pathway, with each intended CoU drawing on the same validation package. For proarrhythmic risk screening, existing electrophysiological data may largely suffice, supplemented by benchmarking against clinically relevant reference compounds. For disease modeling, such as a specific cardiomyopathy, the qualifying body may require additional evidence that this model can recapitulate disease-specific phenotypes.

Adoption facilitation

Qualification is not the end of the process. Test guidelines for qualified NAMs should be published, analogous to those for skin and eye models, enabling consistent adoption. Agencies can also support adoption through stakeholder training, case examples, and decision templates that clarify appropriate uses, evidentiary limits, and reporting expectations. **Table 1** describes how the framework sustains confidence through continued monitoring of qualified NAMs and supports international harmonization over time.

CONCLUSION

The CoU-first model represents progress over ad hoc validation, but it is best suited to simple models with stable endpoints, mature reference data, and limited application domains. For complex NAMs, it requires time- and resource-intensive re-validation of each CoU. As commitments to nonanimal methods deepen across US regulatory practice, validation must evolve so that complex NAMs can deliver the human-relevant evidence they promise. The proposed framework offers a practical path forward: validate the science first, qualify the use second, and maintain confidence through an accessible, iterative evidence base.

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CONFLICT OF INTEREST

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